

Le Nguyen Gia Hung

AI/ML Research Student

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Objective

An AI/ML Research Student with an accepted paper at **AJCAI 2025** for a novel **Transformer** architecture (SmokeNet) in time-series forecasting. Currently developing **Conformer-GAT** fusion models for multimodal speech emotion recognition. Skilled in **PyTorch**, **XAI (SHAP)**, and leading student research initiatives. Seeking a research internship to contribute to complex deep learning challenges.

Education

FPT University

Bachelor of Engineering in Artificial Intelligence

Ho Chi Minh City, Vietnam

Expected Dec 2027

- **Honors:** Certificates of Merit – Honorable Student of Trimester (Semesters 3 & 4).
- **Relevant Coursework:** Data Structures and Algorithms, Linear Algebra, Introduction to Artificial Intelligence.

Skills

- **Programming:** Python, SQL
- **ML/DL:** PyTorch, TensorFlow, Keras, Hugging Face, Scikit-learn, OpenCV
- **Data:** Pandas, NumPy, Matplotlib, SQL (joins, windows)
- **MLOps:** Git, Streamlit, REST API, Weights & Biases
- **Tools:** Linux, Jupyter Notebook, VS Code

Leadership and Activities

Founder & Lead Researcher, SpeedyLabX, FPT University

2025 – Present

- Founded and led a student research group (6 members) focused on applied AI research.
- Spearheaded the Melbourne Air Quality project, resulting in a publication accepted at AJCAI 2025.

Projects

Proactive Air Quality Forecasting and Health Alert System for Melbourne

Academic Research | AJCAI 2025

Accepted

Designed an end-to-end Transformer-based system to forecast $PM_{2.5}$ levels and provide interpretable public health warnings.

- Designed and implemented **SmokeNet (Transformer)** for $PM_{2.5}$ forecasting; achieved $MAE = 0.7470$ and $R^2 = 0.9545$ on the 2024 test set, reducing MAE by **35.9%** vs. the XGBoost baseline.
- Proved robustness in a real-world backtest of the May 2024 severe smoke event, outperforming XGBoost by a substantial **57.7% in MAE**.
- Built end-to-end pipeline: data ingestion from **OpenWeather API** (Nov 2020 – Jan 2025), preprocessing, chronological CV, and GPU training.
- Integrated **XAI** (SHAP, Integrated Gradients, DiCE) for interpretable health guidance and deployed a dashboard prototype.

Enhancing Multimodal Speech Emotion Recognition with Conformer-GAT and XAI

Academic Research | In Progress

Developing a novel Conformer-GAT fusion architecture to model complex audio-text interactions for emotion recognition, with a core focus on Explainable AI (XAI).

- Designing and implementing a novel **Conformer-GAT-Fusion** architecture to integrate sequential acoustic features and graph-based contextual interactions.

- Leveraging **Graph Attention Network (GAT)** to model multimodal (audio-text) dependencies and provide model interpretability.
- Integrating **Explainable AI (XAI)** mechanisms to analyze the model's decision-making process for Speech Emotion Recognition (SER).
- Aiming to benchmark performance on standard datasets (**IEMOCAP**, **RAVDESS**).

Personal Protective Equipment (PPE) Detection System

Course Project DPL302m

*Developed a real-time, multi-class PPE detection system (helmet, vest, gloves) using **YOLOv8** and deployed via a **Streamlit** web interface.*

- Built and trained a **YOLOv8**-based detection pipeline to identify 9 PPE classes (helmet, vest, gloves, boots, and their absence).
- Developed a real-time safety compliance logic to classify workers as "SAFE" or "UNSAFE" based on detected equipment.
- Deployed the system as an interactive **Streamlit** web application with real-time video/webcam processing and results export.

Links: [GitHub](#)

Publications

Le Nguyen Gia Hung *et al.*, "Proactive Air Quality Forecasting and Health Alert System for Melbourne," Accepted at *Australasian Joint Conference on Artificial Intelligence (AJCAI 2025)*.